Name: ______________________ Date: ________________

TRIGONOMETRY, (SINE AND COSINE RULES)

LO: I can use the sine and cosine rules to find sides and angles in non-right-angled triangles.

Sine and cosine rules

The **sine rule**: a/sin A = b/sin B = c/sin C. Used when you know a side and its opposite angle. The **cosine rule**: $a^2 = b^2 + c^2 - 2bc \cos A$.

When to use: Sine rule for ASA/AAS. Cosine rule for SAS (find side) or SSS (find angle).

When to use each

●	Sine rule: know a side-angle pair	a/sin A = b/sin B
●	Cosine rule: know SAS or SSS	$a^2 = b^2 + c^2 - 2bc \cos A$
●	Area = $\frac{1}{2}ab \sin C$	area formula
●	SOH CAH TOA works in all triangles	only right-angled
●	[illegible]	[illegible]

Key idea

Sine rule: a/sin A = b/sin B. Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$.

a/sin A = b/sin B | $a^2 = b^2 + c^2 -$ 2bc cos A

Choose based on what you know

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – sine rule - side

A = 40°, B = 70°, a = 8. Find b.

a/sin A = b/sin B

8/sin 40° = b/sin 70°

b = 8 sin 70°/sin 40° = 11.7

b = 11.7

Cross-multiply to find the unknown.

Example 2 – sine rule - angle

a = 10, b = 7, A = 50°. Find B.

sin B/b = sin A/a

sin B = 7 sin 50°/10 = 0.536

B = $\sin^{-1}(0.536)$ = 32.4°

B = 32.4°

Rearrange to find sin B, then use inverse.

Example 3 – cosine rule - side

b = 8, c = 6, A = 55°. Find a.

$a^2 = 64 + 36 - 2(8)(6)\cos 55°$

= 100 - 55.1 = 44.9

a = 44.9 = 6.7

a = 6.7

Substitute into cosine rule and simplify.

Example 4 – cosine rule - angle

a = 7, b = 9, c = 5. Find A.

cos A = (81 + 25 - 49)/(2 × 9 × 5)

= $\frac{57}{90}$ = 0.633

A = $\cos^{-1}(0.633)$ = 50.7°

A = 50.7°

Rearrange cosine rule to find cos A.

YOUR TURN – PRACTICE (deliberate practice)

1. Sine rule - sides

Find the marked side (1 dp).

a) A=50°, B=60°, a=10. Find b.

b) A=40°, C=80°, a=8. Find c.

c) B=35°, C=75°, b=6. Find c.

d) A=45°, B=65°, a=12. Find b.

2. Sine rule - angles

Find the marked angle (1 dp).

a) a=9, b=7, A=55°. Find B.

b) a=12, c=8, A=70°. Find C.

c) b=10, c=13, B=40°. Find C.

d) a=15, b=11, A=80°. Find B.

3. Cosine rule - sides

Find the marked side (1 dp).

a) b=7, c=9, A=50°. Find a.

b) a=10, c=8, B=60°. Find b.

c) a=6, b=8, C=120°. Find c.

d) b=11, c=14, A=35°. Find a.

4. Cosine rule - angles

Find the marked angle (1 dp).

a) a=5, b=7, c=9. Find C.

b) a=8, b=6, c=10. Find A.

c) a=12, b=9, c=7. Find A.

d) a=4, b=5, c=6. Find C.

5. Mixed

Choose the rule and solve.

a) Sides 8, 10, included angle 45°. Third side?

b) A=38°, B=72°, c=15. Find a.

c) Sides 5, 7, 9. Largest angle?

d) A=110°, a=20, b=12. Find B.

e) b=8, c=11, A=65°. Area?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Sine rule needs a known side-angle pair ☐	**b) Cosine rule can only find sides** ☐	**c) SOH CAH TOA only works in right-angled triangles** ☐	**d) a/sin A = b/sin B is the sine rule** ☐	**e) If you know all 3 sides, use sine rule** ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Two ships leave port. Ship A sails **12 km on 060°**, Ship B sails **8 km on 130°**. Find the distance between them.

Expression: ____________________________

Simplified: ____________________________

b) Triangle sides **a = 10, b = 14, c = 8**. Find all three angles and verify they sum to 180°.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!